

Homework 1

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Course: Stochastic Methods in Mathematical Modelling and AI. AY 2025-2026

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Problem 1. Random Number Generators, 3 points

For producing random numbers one usually uses a pseudo-random number generator (PRNG). A PRNG is an algorithm which delivers a sequence of deterministic numbers that imitate a sequence of random numbers. One of the simplest algorithms is called linear congruential generator (LCG). You are suggested to code and assess its quality. The LCG updates its state X in the following way

$$X_{n+1} = (aX_n + c) \bmod d,$$

where $d > 0$ is named as modulus, a is the multiplier ($0 < a < d$) and c is called as an increment ($0 \leq c < d$). The initial state X_0 is called a seed ($0 \leq X_0 < d$).

1. Implement the LCG with $a = 265$, $c = 133$ and $d = 299$. Generate a sample of 100 numbers from Uniform[0,1] using your LCG seeded with $X_0 = 1$.
2. Numpy default generator is PCG64, a permutation congruential generator developed in 2014. Use `np.random.default_rng(1)` to generate a second sample of 100 numbers from Uniform[0,1].
3. Plot probability density functions and calculate the first 4 moments (mean, variance, skewness and kurtosis) for these samples. Which statistics is closer to the true uniform distribution?
4. Plot autocorrelation functions (ACFs) for LCG and numpy-generated sequences of 100 and 1000 values. Do you observe any changes in the ACFs for longer sequences?

Optionally, apply a https://en.wikipedia.org/wiki/Spectral_test.

Problem 2. Measurable functions and conditional expectation, 2 points

In this exercise, you will get more familiar with formal notions from probability theory: σ -algebras, measurable functions, filtrations, and conditional expectation. We work on the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = ([0, 1], \mathcal{B}([0, 1]), dx)$, where $\mathcal{B}([0, 1])$ is the Borel σ -algebra and dx is the Lebesgue measure on $[0, 1]$. Since $\int_0^1 dx = 1$, this is a probability space.

For example, consider a simple σ -algebra: $\mathcal{G} = \{\emptyset, \Omega, A, \Omega \setminus A\}$ with $0 < \mathbb{P}(A) < 1$. A function g is measurable w.r.t. $\mathcal{G}$ if and only if it is *piecewise constant* on the two sets A and $\Omega \setminus A$:

$$g(x) = c_1 \mathbf{1}_A(x) + c_2 \mathbf{1}_{\Omega \setminus A}(x). \quad (1)$$

In general, the indicator $\mathbf{1}_A(x)$, $A \in \mathcal{F}$, is the simplest example of a measurable function, so we will construct measurable functions by a linear combination of indicators.

A *filtration* is a growing sequence of σ -algebras $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \dots \subset \mathcal{F}_n \subset \dots \subset \mathcal{F}$. For example, the following is a filtration: $\{\emptyset, \Omega\} \subset \{\emptyset, \Omega, A, \Omega \setminus A\} \subset \{\emptyset, \Omega, A, B, C, A \sqcup B, A \sqcup C, B \sqcup C\}$, where in the last σ -algebra $\Omega = A \sqcup B \sqcup C$ and $B \sqcup C = \Omega \setminus A$.

For a smaller σ -algebra $\mathcal{G} \subset \mathcal{F}$, the *conditional expectation* $\mathbb{E}[f | \mathcal{G}]$ of an $\mathcal{F}$ -measurable function f is the best (mean-squared-error) approximation of f by functions measurable with respect to $\mathcal{G}$.

For $n \in \mathbb{N}$, define the level- n *dyadic partition* of $[0, 1]$ by the half-open intervals

$$I_k^{(n)} = [k2^{-n}, (k+1)2^{-n}), \quad k = 0, 1, \dots, 2^n - 1. \quad (2)$$

Explicitly, we get the following partitions:

$$\begin{aligned} n = 1 : & \quad [0, \frac{1}{2}), [\frac{1}{2}, 1], \\ n = 2 : & \quad [0, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}), [\frac{1}{2}, \frac{3}{4}), [\frac{3}{4}, 1], \text{ etc.} \end{aligned}$$

Let $\mathcal{F}_n$ be the σ -algebra generated by the level- n dyadic partition, i.e., $\mathcal{F}_n = \sigma(\{I_k^{(n)}\}_{k=0}^{2^n-1})$. For example, $\mathcal{F}_1 = \sigma(\{[0, \frac{1}{2}), [\frac{1}{2}, 1]\}) = \{\emptyset, [0, \frac{1}{2}), [\frac{1}{2}, 1], [0, 1]\}$.

Answer the following questions:

1. Show that $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$ is indeed a filtration. Find an explicit analytical formula for all functions measurable with respect to the sigma algebra $\mathcal{F}_n$. All such functions form a finite-dimensional vector space; what is the dimension of this space (i. e. number of trainable weights)?
2. Let $f(x) = \cos(\pi x)$ on $[0, 1]$.
 - (a) Use the MSE-optimality to derive an explicit formula for the conditional expectation of cosine with respect to a dyadic sigma algebra, $\mathbb{E}[f | \mathcal{F}_n]$. Show that the bigger our sigma algebra $\mathcal{F}_n$, the lower MSE we will get (plot MSE as a function of n).
 - (b) Compute and plot the first four conditional expectations $\mathbb{E}[f | \mathcal{F}_n]$ for $n = 0, 1, 2, 3$.
 - (c) Verify that, for every n , $\int_0^1 \mathbb{E}[f | \mathcal{F}_n](x) dx = \int_0^1 f(x) dx$ (preservation of the mean).

Problem 3. Sum of Random Variables, 2 points

Independent random variables X_1 and X_2 follow the Cauchy distribution

$$p(x) = \frac{1}{\pi (1 + (x - 2)^2)}$$

1. Find the distribution of their sum $Y = X_1 + X_2$
2. Sample 40000 values for both X_1 and X_2 . Plot a histogram for X_1, X_2 and Y
3. Check the correspondence between the histogram and the analytical solution (histograms can be displayed in the interval $[-25, 25]$ for a convenience).
4. Can we say that Y follows a stable distribution?

Problem 4. Correlated Sampling, 3 points

In this problem we are interested in two sets of dependent random variables. Imagine set X_1 as the price of a product and set X_2 as the price of the constituent materials of a product. Obviously when the price of a constituent material is high, we expect that the final price of that specific product will be high as well.

1. Design a sampling procedure that provides two sets of correlated random variables, uniformly distributed in $[0, 1]$.
2. Gradually change correlation coefficient $r = 0, 0.6, 0.9$ and plot the joint distribution for x_1 and x_2 . Additionally plot marginal distributions for x_1, x_2 and check that they are uniform.